

# PROJECT TRICORDER MK. I: SUBSPACE AUTOMATIC GATEWAY TECHNICAL REPORT

Asymmetric 5G/Wi-Fi to Meshtastic/LoRa Operational Analysis

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Prepared for Field Operations

October 2026

## 1 Executive Summary

This report presents a technical and operational evaluation of the Project Tricorder Mk. I Subspace gateway architecture, examining the asynchronous bridging of internet-backed data services (5G/Wi-Fi API endpoints) across an off-grid Meshtastic LoRa mesh network. Utilizing field test logs captured between dual field nodes—designated **Kirk** (CC118D, gateway-enabled) and **Spock** (385FD9, off-grid RF node)—this analysis details packet serialization, fragment truncation behavior, distance-based attenuation, and multi-transmit buffer re-assembly over a 1-mile mobile propagation vector.

## 2 System Architecture & Asymmetric Gateway Topology

The Project Tricorder platform implements an asymmetric telemetry bridge where network capabilities are divided by node role:

- **The Gateway Node (Kirk - CC118D):** Equipped with active cellular (5G) and internet-backed socket connectivity, Kirk continuously polls external civic and flight tracking APIs, serializes JSON payloads into compact subspace frame types (TF), and broadcasts them over the local 915 MHz LoRa mesh.
- **The Off-Grid Node (Spock - 385FD9):** Operating strictly via Meshtastic/LoRa transport without direct internet access, Spock acts as a mobile remote client transmitting manual Push-to-Talk (PTT) strings, position beacons (POS), and aloha heartbeats (AH) back toward the gateway.
- **Protocol & Reassembly Layer:** The custom sub-space transport layer utilizes connectionless UDP-style multi-transmits, NRZI streaming, and frame-type prefixes. Because extended LoRa propagation paths over variable terrain frequently encounter window constraints, partial packet fragments are caught by local buffers and automatically reassembled into 100% complete text payloads.

## 3 Comprehensive Transmission & Reception Master Logs

The following tables synthesize verified exchanges between Kirk and Spock extracted from the field test logs, detailing directionality, timestamps, raw payloads, positional coordinates, calculated separation distances, and reassembly delivery status.

### 3.1 Spock Uplink Messages (Spock → Kirk)

| Send Time | Sender Pos.            | Message Content                               | Con- | Receiver Pos.          | Dist.  | Status                                 |
|-----------|------------------------|-----------------------------------------------|------|------------------------|--------|----------------------------------------|
| 18:37:52  | 41.52553,<br>-74.08764 | test test<br>tricorder test                   |      | 41.52662,<br>-74.08900 | ≈150 m | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:44:45  | 41.52428,<br>-74.08184 | Captain I<br>require your<br>help             |      | 41.52642,<br>-74.08894 | ≈610 m | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:52:10  | 41.53294,<br>-74.08821 | Captain Kirk<br>can you hear me               |      | 41.52642,<br>-74.08894 | ≈725 m | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:53:29  | 41.52575,<br>-74.08703 | This message<br>is from a<br>Tricorder        |      | 41.52642,<br>-74.08894 | ≈180 m | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:55:53  | 41.53193,<br>-74.08968 | Hello this<br>is Mr Spock's<br>Tricorder      |      | 41.52642,<br>-74.08894 | ≈612 m | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 19:01:16  | 41.52523,<br>-74.08369 | Captain Kirk<br>please send a<br>medical team |      | 41.52642,<br>-74.08894 | ≈470 m | <b>FULLY RECEIVED</b><br>(Reassembled) |

### 3.2 Kirk Gateway API Event Broadcasts (Kirk → Spock)

| Send Time | Sender Pos.            | Event & Payload Content                                          | Receiver Pos.          | Dist.   | Status                                 |
|-----------|------------------------|------------------------------------------------------------------|------------------------|---------|----------------------------------------|
| 18:35:02  | 41.52680,<br>-74.08897 | <b>Civic:</b> Potholes,<br>Other Street Repair<br>— Hawthorne Dr | 41.52657,<br>-74.08906 | ≈25 m   | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:44:04  | 41.527,<br>-74.089     | <b>Flight:</b> ICE623<br>2621 m 132 m/s<br>KEF to EWR            | 41.52417,<br>-74.08128 | ≈670 m  | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:45:05  | 41.527,<br>-74.089     | <b>Flight:</b> ICE623<br>2621 m 132 m/s<br>KEF to EWR            | 41.52417,<br>-74.08128 | ≈670 m  | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:46:05  | 41.527,<br>-74.089     | <b>Flight:</b> POE607<br>3749 m 173 m/s<br>YYZ to LGA            | 41.52417,<br>-74.08128 | ≈670 m  | <b>FULLY RECEIVED</b><br>(Reassembled) |
| 18:47:05  | 41.527,<br>-74.089     | <b>Flight:</b> POE607<br>3749 m 173 m/s<br>YYZ to LGA            | 41.53502,<br>-74.07279 | ≈1.2 km | <b>FULLY RECEIVED</b><br>(Reassembled) |

## 4 Link Performance & Engineering Findings

1. **Zero-Loss Reassembly Efficiency:** Across all evaluated transmissions—ranging from short 150-meter line-of-sight checks to extended 725-meter marginal fringe tests—**100% of messages** were ultimately reassembled successfully by the receiving node protocol layer.
2. **Mitigation of RF Truncation:** Atmospheric attenuation and multipath interference in suburban tree/structure cover caused initial packet arrivals to truncate into short prefixes. However, the sub-space protocol's sliding buffer window and redundant burst design completely eliminated permanent data loss.
3. **Asymmetric Viability:** The test validates that an internet-tethered gateway node (Kirk) can successfully ingest, translate, and push external data streams (aviation radar and municipal advisories) to sovereign, off-grid field instruments (Spock) over standard 915 MHz LoRa hardware.

## 5 Subspace Protocol Overview

The **Project Tricorder Subspace Protocol** is engineered for sovereign, low-bandwidth, resilient edge-to-edge communication over Meshtastic/LoRa links. Its core architectural mechanics include:

- **Compact Tuple & Frame Header Structure:** Replaces verbose protocol overhead with deterministic byte sequences and prefix tokens (AH for Aloha heartbeats, POS for coordinate broadcasts, PTT for intercom streams, and TF for telemetry/gateway frames).
- **Connectionless UDP Multi-Transmit Model:** Bypasses heavy TCP handshake states in favor of deterministic, repeating frame broadcasts with local de-duplication.
- **Alternating Parity & Burst Error Correction:** Employs dual-stage redundancy and rolling buffer windows to reconstruct truncated string fragments automatically on the receiving node, ensuring data integrity across adverse propagation environments.